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Tally Ho! – Explore the Varied Uses of Tally Tables

SQLRV, LLC
HEALTHCARE INFORMATICS
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Just who is this guy?

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40 Years

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Agenda

- What are Tally Tables?
  - Purposes and Uses
  - How To Create Them
- How to Use
  - Find Missing Data
  - Break Strings
  - Generate Data
  - Replace Loops & Cursors
- Closing Thoughts

5 Basic Concepts of Any Programming Language

1. Program Structure
   - The overall form of a program
   - Interrelationships between individual components
2. Variable Declaration (Names & Data Types)
3. Looping Structures (Conditional)
4. Control Structures (Logic flows & branches)
5. Syntax (Language Rules)
   - ***** Nothing About Set-Based Operations *****

Procedural vs Set-Based

Procedural Programming

- Data is very abstract
  - Structure or organization of the data is unimportant
  - Separated by different layers of abstraction
- Concerned with what to do and how to do it (Imperative)
  - Step-by-step
  - Statements, Loops, Object manipulation, etc.
  - Different logic paths depending upon data attributes
  - Record by record processing
- Provides total control
  - You know exactly what it will do
  - Execution method will be consistent each and every time

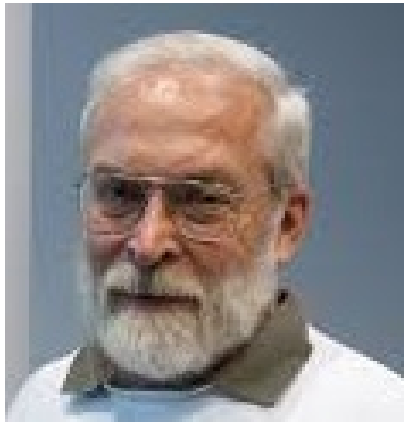
Set-Based Programming

- Describes what should be done (Declarative)
  - Indicates the operation
  - Identifies the conditions
- Allows the engine to determine an optimal method to obtain results based upon:
  - Data structure, organization and volume
  - Statistics from previous runs
- Frees developer from mechanical/specific details
- Allows for changing conditions without altering query
- RDBMS is optimized to select or manipulate records sets as a whole (often at the expense of procedural operations)

What Are Tally Tables?

What Are Tally Tables?

- A numerical sequence table
- Perform set-based sequence logic
  - Replace loops
  - Improve performance
- Jeff Moden
  - RBAR -“Row By Agonizing Row”
  - Avoid Processing Data Each Row at a Time
  - Most common performance killer in T-SQL
- Promote Set-Based Operations



How Does It Work?

Results | Message

	StringValue
1	One
2	Two
3	Three
4	Four
5	Five

+

Results

	n
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10

=

Results | Messages

	n	StringValue
1	1	>O<ne
2	2	O>n<e
3	3	On>e<
4	1	>T<wo
5	2	T>w<o
6	3	Tw>o<
7	1	>T<hree
8	2	T>h<ree
9	3	Th>r<ee
10	4	Thr>e<e
11	5	Thre>e<
12	1	>F<our
13	2	F>o<ur
14	3	Fo>u<r
15	4	Fou>r<
16	1	>F<ive

LIVE
DEMO

Set-Based Processing

- Does NOT mean "all in one query“
  - A single query can contain a hidden loop
- Does NOT mean something without a loop
- CAN mean something that has a loop
  - Certain queries require multiple SETS of information to be processed
- Does mean "touching" each row minimally
- Does require a paradigm shift in thinking

Other “Loops” and “Cursors”

- Recursion
  - A recursive CTE (rCTE) calls itself as a hidden loop
  - Can kill performance with about 3 times (or more) the resources of a simple while-loop
  - Most rCTEs can be re-written in a set-based manner using a simple while-loop
    - rCTE loops through each level of recursion
    - Each loop requires the results of the previous loop
    - Performing loop explicitly save resources to run faster
    - Example: 3 levels of recursion can be processed in only 3 loops

Other “Loops” and “Cursors”

- Explicit Cursor
  - Query places data in an internal structure
  - Fetches record one at a time
  - Default settings allow for forward-backward traversing and data updates
  - Consumes extensive resources
  - If required use LOCAL, FAST_FORWARD, and READ_ONLY options

Other “Loops” and “Cursors”

- Internal Loops
  - The hidden but very high speed looping effect that set-based code experiences behind the scenes
  - A simple SELECT "iterates" through rows but in a manner that the optimizer knows best
  - Example: Windowing functions

What We Covered

- Purposes and Uses
- How To Create Them
  - Inline Table-Valued Function
  - Fixed Table
- How to Use
  - Find Missing Data
  - Perform String Splits
  - Generate Data
  - Replace Many Types of Loops & Cursors

Closing Thoughts

- Benefits over loops & cursors:
  - Set-Based
    - Improves performance
    - Simplifies the amount of code needed
  - Improved Maintainability
    - More concise code
    - Less chance of errors
  - Check out SQLServerCentral.com for more

References

- Jeff Moden: http://www.sqlservercentral.com/articles/T-SQL/62867/
- Dwain Camps: http://www.sqlservercentral.com/blogs/dwainsql/2014/03/27/tally-tables-in-t-sql/
- The Code Project: https://www.codeproject.com/Articles/712738/SQL-Wizardry-Part-Eight-Tally-Tables
- Red Gate: https://www.red-gate.com/simple-talk/sql/database-administration/creative-solutions-by-using-a-number-table/

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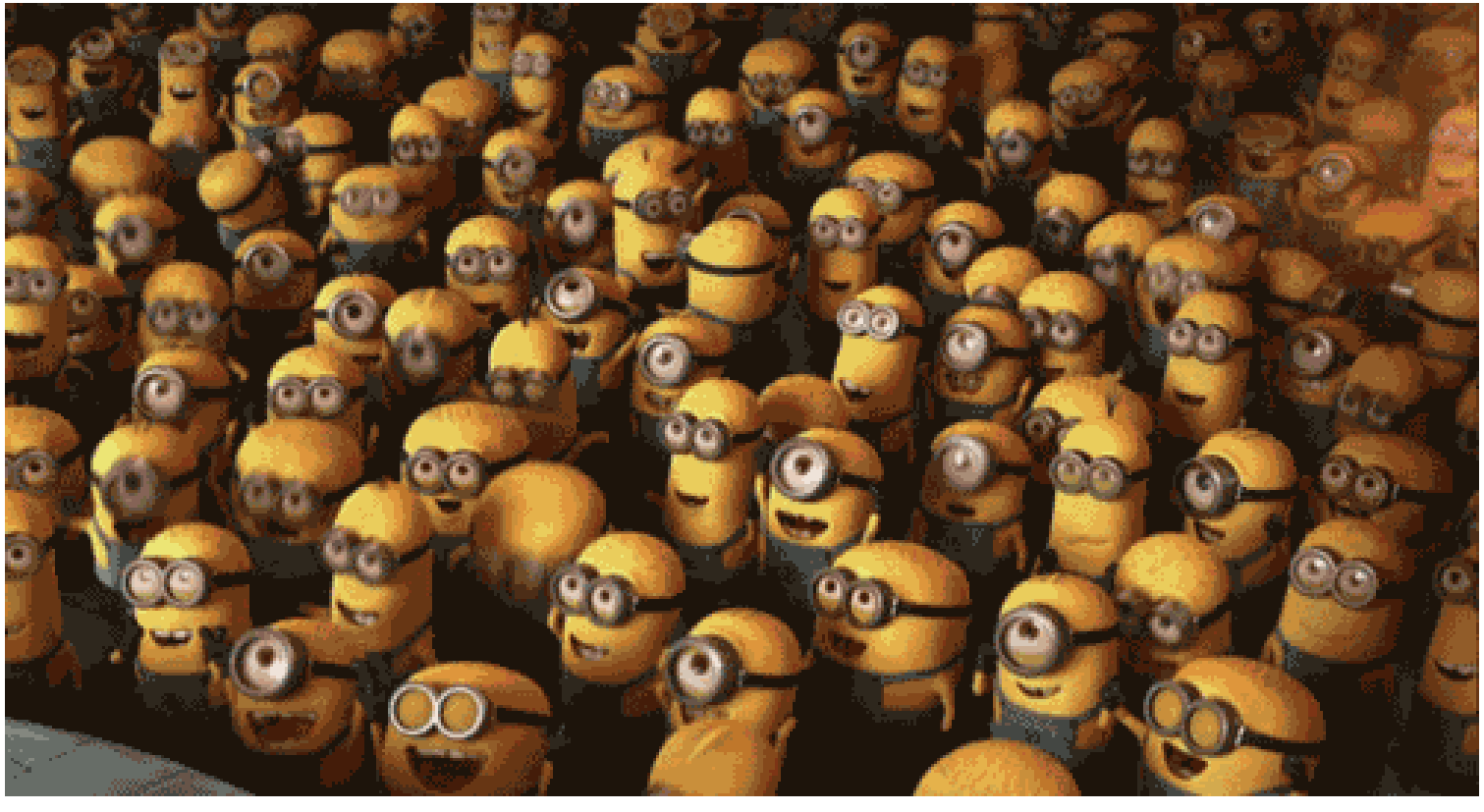

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